A coloring-free approach to materialising Jacobians and Hessians in JAX by leveraging sparsity

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Abstract

Automatic differentiation (AD) has proven indispensable in powering the ML revolution by enabling accelerated gradient-based learning without hard-to-maintain hand-coded analytic derivatives. However, modern AD libraries such as JAX still leave significant room for improvement when it comes to computation of full Jacobians and Hessians which are central to root finding (e.g. in implicit ODE solvers or deep equilibrium models) or second-order constrained optimisation (relevant to robotic and optimal control). In this work, we show that leveraging JAX's functional transformations and introducing new sparse formats, including batched Ellpack, can reduce redundant computation and exploit internal structure achieving orders of magnitude speedups in structured nonlinear problems (including the CUTEst suite). This functionality is made available in an open source library with a drop-in API mirroring JAX's (including hessian, jacfwd and jacrev with a choice of output formmats). Our methodology is competitive with colouring methods and outperforms it on some problems without requiring the user to manually provide the sparsity pattern and avoiding the sometimes expensive coloring phase. Furthermore, when the sparsity pattern is statically known it can be obtained for free at trace time. We envisage the key use case to be coupling the accelerated Jacobian/Hessian computation to a sparse solver such as CudSS.

1 Introduction

The JAX library provides a powerful framework for implementing large ML models/simulators and training/optimizing them using gradient-based methods (e.g. Adam, L-BFGS) on modern accelerators. For these cases, the computational efficiency of the gradient as obtained by JAX's automatic differentiation will usually be competitive with any handcoded version, greatly reducing development time and improving maintainability. However, the same cannot be said for the case of root finding (e.g. in implicit ODE solvers or deep equilibrium models) and second-order optimisation using direct solvers (relevant to robotics and optimal control) where the entire Jacobian and Hessian must be obtained. This is because JAX computes these simply by applying the jvp/hvp to a batch of basis vectors rather than leveraging structural sparsity; the resulting $\mathcal{O}(n^2)$ cost of computing the Jacobian/Hessian is most acute when the matrices possess structure (e.g. sparsity) that enables Traditional ways . function transformation

2 State of the art

sectionMethodology

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Operation	Dense	values	in_cols	start	end
(seed) $x = \texttt{Identity}(5)$	$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$	$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$	0	5
$a = c \odot x, c = (3, 5, 7, 9, 11)$	$\begin{pmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 & 0 \\ 0 & 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 0 & 11 \end{pmatrix}$	$\begin{bmatrix} 3 \\ 5 \\ 7 \\ 9 \\ 11 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$	0	5
$b = a_{1:5}$	$\begin{pmatrix} 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 & 0 \\ 0 & 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 0 & 11 \end{pmatrix}$	$\begin{bmatrix} 5 \\ 7 \\ 9 \\ 11 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$	0	4
$d = a_{0:4}$	$\begin{pmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 & 0 \\ 0 & 0 & 0 & 9 & 0 \end{pmatrix}$	$\begin{bmatrix} 3 \\ 5 \\ 7 \\ 9 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$	0	4
$e = b - d$	$\begin{pmatrix} -3 & 5 & 0 & 0 & 0 \\ 0 & -5 & 7 & 0 & 0 \\ 0 & 0 & -7 & 9 & 0 \\ 0 & 0 & 0 & -9 & 11 \end{pmatrix}$	$\begin{bmatrix} -3 & 5 \\ -5 & 7 \\ -7 & 9 \\ -9 & 11 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$	0	4
$f = \texttt{pad}(e,\ 1 \to 0)$	$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ -3 & 5 & 0 & 0 & 0 \\ 0 & -5 & 7 & 0 & 0 \\ 0 & 0 & -7 & 9 & 0 \\ 0 & 0 & 0 & -9 & 11 \end{pmatrix}$	$\begin{bmatrix} -3 & 5 \\ -5 & 7 \\ -7 & 9 \\ -9 & 11 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$	1	5

Table 1: Walk for $f(x) = \mathrm{pad}\big((c \odot x)_{1:} - (c \odot x)_{:-1}\big)$ with $c = (3, 5, 7, 9, 11)$, $\mathrm{len}(x) = 5$. The mathematical Jacobian (parentheses, right-aligned for sign clarity) is contrasted with the BEllpack representation (square brackets, the literal in-memory tensors). Elementwise scaling rewrites `values` in place; slicing shifts the columnfband; subtraction concatenates two k=1 bands into a k=2 band; pad updates only `start`/`end`.

2.1 Sparse representations

2.2 You only linearize once

3 Experiments

3.1 Nonlinear Heat Equation

$$\frac{partialT_i}{\partial t} = \nabla \kappa \nabla T_i = (\kappa_{i+1/2}) \tag{1}$$

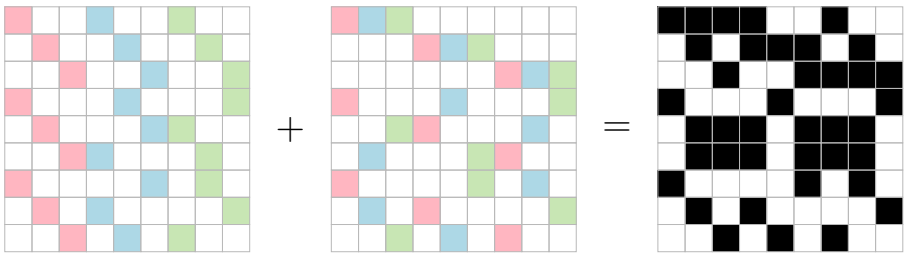
3.2 CUTEst problems

3.3 Pole-cart collocation

Identity → Constant Diagonal → Diagonal → BEllpack → BCOO

Table 2: Categorised list of linear JAX primitives supported by Ellpax (trailing `_p` dropped).

Group	Primitive
Shape-preserving unary	`conj, copy, imag, neg, real, cumsum, convert_element_type`[1]
Shape-affecting unary	`pad, reshape, broadcast_in_dim, reduce_sum, squeeze`
Indexing	`slice, rev, gather, transpose, scatter_add`
Shape-preserving multilinear	`add`[2]`, sub, mul, div`
Shape-affecting multilinear	`dot_general, concatenate`
Control flow	`cond, select_n, (jit)`
Multi-output unary	`split`



(a) $J(g_1)$, 3-colouring by i (b) $J(g_2)$, 3-colouring by j (c) $J(g_1+g_2)$, no $k<9$ colouring

Figure 1: Boolean Jacobian patterns at $P = 3$. Each summand is 9×9 with 3 nonzeros per row; the column-conflict graph of $J(g_1)$ admits a 3-colouring grouped by row index i (panel a, columns of grid points sharing i never co-occur in a g_1-line), and likewise $J(g_2)$ admits a 3-colouring by j (panel b, the transverse partition). When the two Jacobians are added, every pair of distinct columns is joined by some g_1- or g_2-line, so the aggregate column-conflict graph is complete and no compression below 9 colours exists (panel c, shown without colouring).

4 Coloring-pathologic projective planes

Many of the CUTEst problems involve objective functions of the form:

$$f(x) = \sum_i f_i(x), \tag{2}$$

where

5 Bib test

This section exercises the bibliography wiring; it should be removed before submission.

Single citation: foundational sparse Jacobian estimation [7]. Two-cite: graph-theoretic formulation [4, 5]. Three consecutive (compressed): coloring lineage [7, 18, 4]. Survey reference [11]; recent comparator work [15, 8].

Partial-separability [13, 6]; multifrontal-assembly analogy [9, 16]. Experiments on the CUTEst sweep [12, 2], bit-exact against the Pearlmutter HVP oracle [17]. Background AD and sparse storage [14, 1, 19]; host system [10, 3].

6 Conclusions

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A Technical appendices and supplementary material

Technical appendices with additional results, figures, graphs, and proofs may be submitted with the paper submission before the full submission deadline (see above). You can upload a ZIP file for videos or code, but do not upload a separate PDF file for the appendix. There is no page limit for the technical appendices.

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B Action of primitive rules on a BEllpack operand

We use the 5-tuple shorthand $\mathrm{BE}[V,\, C,\, s,\, e,\, T]$ for a BEllpack with values V, in-column tuple C of length k, row range $[s, e)$, and transpose flag T. The remaining fields (`out_size`, `in_size`, `batch_shape`) pass through unchanged unless explicitly modified. Each row below is the canonical structural case implemented in the current source; rules with several distinct branches are listed once per branch with a labelled left-hand side. When the listed branch does not apply, the rule densifies via $V := \mathrm{BE.todense}$ or promotes to BCOO; those fall-back paths are not enumerated here.

Operation	Result
Shape-preserving unaries	
$\Phi(\mathrm{BE})$[3]	$\mathrm{BE}[\Phi(V),\, C,\, s,\, e,\, T]$
`cumsum`(BE)	`cumsum`(BE.todense) (no structural path)
Shape-affecting unaries	
`pad`(BE, before, after), no interior	$\mathrm{BE}[V,\, C,\, s + \mathrm{before},\, e + \mathrm{before},\, T]$
`slice`(BE, $[\ell, h)$), unit stride on out-axis	$\mathrm{BE}[V_{[\ell:h]},\, C_{[\ell:h]},\, s - \ell,\, e - \ell,\, T]$
`squeeze`(BE), `out_size`=1 at s=0, e=1	$\mathrm{BE}[V,\, C,\, s,\, e,\, T]$ (identity)

[3] Φ stands for any element-wise unary that preserves shape and the zero element: $\Phi \in \{$`neg`, `copy`, `conj`, `real`, `imag`, `convert_element_type`$\}$. Each maps $V \mapsto \Phi(V)$ pointwise on the values tensor, leaving C, s, e, T untouched.

Operation	Result
`reshape`(BE), batched→unbatched, T=False	BE[V_{flat}, C_{tile}, 0, $\prod b \cdot$ `nrows`, T]
`reshape`(BE), unbatched, leading singleton	BE[V_{reshape}, C_{tile}, s, e, T], b'=$(1, \dots, 1)$
`broadcast_in_dim`(BE), 1-row T=True tile to N	BE[V, broadcast(C, N), 0, N, True]
`broadcast_in_dim`(BE), leading-batch insert	BE[V_{bcast}, C_{bcast}, s, e, T], b'=(new_batch, b)
`reduce_sum`(BE), axis=out, static C	BE[V_{Σ}, dedup(C), 0, 1, T]
`reduce_sum`(BE), axis⊆batch	BE[V.sum(axes), C, s, e, T], b'=$b \setminus$ axes
Indexing	
`transpose`(BE, perm), 2×2 cross-V swap	BE[V, C, s, e, $\neg T$]
`transpose`(BE, perm), batch-only / V-fixed	BE[V_{perm}, C_{perm}, s, e, T]
`rev`(BE), axis=out, unbatched	BE[`flip`(V), `flip`(C), $N-e$, $N-s$, T]
`gather`(BE, ...)	raises `NotImplementedError` (densify upstream)
`scatter_add`(·, BE, idx), kept form	`scatter_add_op`(V, idx, ...) via dispatch
Shape-preserving multilinear	
$\alpha \cdot$ BE, scalar α	BE[$\alpha\, V$, C, s, e, T]
$\mathbf{w} \odot$ BE, per-row weight broadcast	BE[$\mathbf{w}\, V$, broadcast(C), s, e, T]
BE$/\alpha$, scalar α	BE[V/α, C, s, e, T]
$\text{BE}_1 + \text{BE}_2$, identical (s, e, b, C)	BE[$V_1 + V_2$, C, s, e, T]
$\text{BE}_1 + \text{BE}_2$, identical (s, e, b), mixed C	dedup(BE[$V_1 \parallel V_2$, $C_1 \parallel C_2$, s, e, T])
$\text{BE}_1 + \text{BE}_2$, row-disjoint, abutting	BE[$V_1 \parallel V_2$, $C_1 \parallel C_2$, min(s_i), max(e_i), T]
BE + BCOO (or any incompatible)	BCOO(BE) + BCOO via shared concat path
BE − BE′	BE + (−BE′) via the add path
Shape-affecting multilinear	
BE $\cdot\, M$, M closure 2-D, contract on in-axis	dense V_{contract} of shape (b, N, B) where B=M.shape[1]
`concatenate`$(\text{BE}_1, \text{BE}_2)$, axis$< n_{\text{batch}}$	BE[$V_1 \parallel_{\text{ax}} V_2$, C_{widen}, s, e, T]
`concatenate`$(\text{BE}_1, \text{BE}_2)$, axis=out	BE[$V_1 \parallel V_2$, $C_1 \parallel C_2$, 0, $N_1 + N_2$, T]
Control flow / selection	
`cond`(pred, ...), pred static	recurse into chosen branch jaxpr
`jit`(BE)	recurse into inner jaxpr; outputs marked traced
`select_n`(p, BE, closures), single traced	BE[mask$_p \cdot V$, C, s, e, T]
`select_n`(p, BE_1, BE_2), identical (s, e, b, C)	BE[`select_n`(p, V_1, V_2), C, s, e, T]
Multi-output unary	
`split`(BE), batched, axis=out	list of BE[$V_{[c_i]}$, $C_{[c_i]}$, 0, $e_i - s_i$, T]
`split`(BE), unbatched, static C	list mixing BE chunks and empty BCOO chunks

The rules above cover the BEllpack-preserving paths in `lineaxpr/_linops/ellpack_transforms.py`, `lineaxpr/_linops/ellpack.py`, and `lineaxpr/_rules/{add,mul,multilinear,structural,control_flow}.py`. Several primitives (`reshape`, `broadcast_in_dim`, `reduce_sum`) carry additional structural branches for less common shape combinations; we have shown the canonical case for each and refer the reader to the cited files for the full case-split.

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